

Continuum Mechanics

Steven M. Wise

swise1@utk.edu
University of Tennessee

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Chapter 05

Balance Laws
Part 4



Local Lagrangian Form of the Second Law



Result 5.19 (Local Lagrangian Second Law)

Let

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

be a motion of a continuum body. Then the Second Law of Thermodynamics implies

$$\rho_0 \dot{\eta}_m = \rho_0 \frac{R}{\Theta} - \nabla^{\mathsf{X}} \cdot \left(\frac{\mathbf{Q}}{\Theta} \right) + \sigma_m(\mathbf{X}, t) \det F(\mathbf{X}, t), \qquad \mathbf{X} \in B, \quad t > 0,$$

where

$$\sigma_m(\mathbf{X}, t) \det F(\mathbf{X}, t) \geq 0, \qquad \mathbf{X} \in B, \quad t > 0.$$

Result 5.19 (Cont.)

The material temperature and heat-supply fields are

$$\Theta(\mathbf{X}, t) := \theta_m(\mathbf{X}, t) = \theta\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big),$$

$$R(\mathbf{X}, t) := r_m(\mathbf{X}, t) = r\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big).$$

The material heat-flux field is

$$\mathbf{Q}(\mathbf{X}, t) := \det F(\mathbf{X}, t) F(\mathbf{X}, t)^{-1} \mathbf{q}_m(\mathbf{X}, t).$$

Here σ_m is the material description of the spatial entropy-production density:

$$\sigma_m(\mathbf{X}, t) := \sigma\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big) \geq 0.$$



Proof.

Axiom 5.4, the Second Law of Thermodynamics, states that

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{\Omega_t}\rho\eta\,\mathrm{d}V_{\mathrm{x}} = \int_{\Omega_t}\frac{\rho r}{\theta}\,\mathrm{d}V_{\mathrm{x}} - \int_{\partial\Omega_t}\frac{\mathbf{q}\cdot\mathbf{n}}{\theta}\,\mathrm{d}A_{\mathrm{x}} + \int_{\Omega_t}\sigma\,\mathrm{d}V_{\mathrm{x}}$$

for every $\Omega_t \subseteq B_t$.

Changing variables and using conservation of mass,

$$\begin{aligned}\int_{\Omega_t}\rho\eta\,\mathrm{d}V_{\mathrm{x}} &= \int_{\Omega_0}\rho_m\eta_m\det F\,\mathrm{d}V_{\mathrm{X}}\\ &= \int_{\Omega_0}\rho_0(\mathbf{X})\eta_m(\mathbf{X},t)\,\mathrm{d}V_{\mathrm{X}},\end{aligned}$$

because $\rho_m \det F = \rho_0$.

Proof (Cont.)

Similarly,

$$\int_{\Omega_t} \frac{\rho r}{\theta}\, \mathrm{d}V_x = \int_{\Omega_0} \frac{\rho_m r_m}{\theta_m} \det F\, \mathrm{d}V_X$$
$$= \int_{\Omega_0} \rho_0(\mathbf{X}) \frac{R(\mathbf{X}, t)}{\Theta(\mathbf{X}, t)}\, \mathrm{d}V_X.$$

Changing variables and using the definition of σ_m,

$$\int_{\Omega_t} \sigma\, \mathrm{d}V_x = \int_{\Omega_0} \sigma_m(\mathbf{X}, t) \det F(\mathbf{X}, t)\, \mathrm{d}V_X.$$

Proof (Cont.)

Finally, Result 4.11 and the Divergence Theorem give

$$\begin{aligned}\int_{\partial\Omega_t} \frac{\mathbf{q}\cdot\mathbf{n}}{\theta}\,\mathrm{d}A_{\mathrm{x}} &= \int_{\partial\Omega_0} \frac{\mathbf{q}_m}{\theta_m}\cdot\left(\det F\, F^{-\mathrm{T}}\mathbf{N}\right)\,\mathrm{d}A_{\mathrm{X}} \\ &= \int_{\partial\Omega_0}\left(\frac{\mathbf{Q}}{\Theta}\right)\cdot\mathbf{N}\,\mathrm{d}A_{\mathrm{X}} \\ &= \int_{\Omega_0}\nabla^{\mathrm{X}}\cdot\left(\frac{\mathbf{Q}}{\Theta}\right)\,\mathrm{d}V_{\mathrm{X}}.\end{aligned}$$

Proof (Cont.)

Thus the Second Law may be expressed as

$$\int_{\Omega_0} \rho_0 \dot{\eta}_m \, \mathrm{d}V_X = \int_{\Omega_0} \left\{ \rho_0 \frac{R}{\Theta} - \nabla^X \cdot \left(\frac{\mathbf{Q}}{\Theta} \right) + \sigma_m \det(F) \right\} \mathrm{d}V_X.$$

Since $\Omega_0 \subseteq B$ is arbitrary, the Localization Theorem gives

$$\rho_0 \dot{\eta}_m = \rho_0 \frac{R}{\Theta} - \nabla^X \cdot \left(\frac{\mathbf{Q}}{\Theta} \right) + \sigma_m \det(F).$$

Because $\sigma_m \geq 0$ and $\det F > 0$, the product $\sigma_m \det F$ is nonnegative. □

Result 5.20 (Reduced Clausius–Duhem Inequality)

Let $\boldsymbol{\phi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body. Then the Second Law of Thermodynamics implies that

$$\rho_0 \dot{\Psi} = P : \dot{F} - \rho_0 \eta_m \dot{\Theta} - \frac{\mathbf{Q}}{\Theta} \cdot \nabla^X \Theta - \Theta \sigma_m \det(F),$$

for every $\mathbf{X} \in B$ and $t > 0$, where

$$\Theta \sigma_m \det(F) \geq 0 \qquad \text{and} \qquad \Psi(\mathbf{X}, t) := \Phi(\mathbf{X}, t) - \Theta(\mathbf{X}, t) \eta_m(\mathbf{X}, t).$$

Proof.

Exercise. □



Frame Indifference

Definition (5.21 (Superposed Rigid Motion))

Two motions

$$\boldsymbol{\phi}, \boldsymbol{\phi}^* : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

of a body are related by a *superposed rigid motion* if, for every $\mathbf{X} \in B$ and $t \geq 0$,

$$\boldsymbol{\phi}^*(\mathbf{X}, t) = Q(t)\boldsymbol{\phi}(\mathbf{X}, t) + \mathbf{c}(t), \tag{5.36}$$

for some rotation $Q(t)$ and vector $\mathbf{c}(t)$.

For every $t \geq 0$,

$$Q(t)^{\mathsf{T}} Q(t) = Q(t) Q(t)^{\mathsf{T}} = I, \qquad \det Q(t) = 1.$$



Different pairs of observers are described by different functions $Q(t)$ and $\mathbf{c}(t)$. We assume that the observers use the same synchronized clocks.

For the corresponding spatial points, define

$$\mathbf{x}^* = \mathbf{g}_t(\mathbf{x}) = \mathbf{g}(\mathbf{x}, t) := Q(t)\mathbf{x} + \mathbf{c}(t), \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$

Thus the two observers assign the coordinates $\mathbf{x}$ and $\mathbf{x}^*$ to the same material particle at time t.



Result 5.22 (Kinematic Transformations)

Let $\boldsymbol{\phi}$ and $\boldsymbol{\phi}^*$ be two motions related by equation (5.36). Then their material fields satisfy

$$F^* = QF, \qquad R^* = QR,$$

$$U^* = U, \qquad V^* = QVQ^{\mathsf{T}}, \qquad C^* = C,$$

for every $\mathbf{X} \in B$ and $t \geq 0$.

Result 5.22 (Cont.)

The spatial velocity gradients are related by

$$\nabla^{x^*}\mathbf{v}^*(\mathbf{x}^*,t)\Big|_{x^*=g(x,t)} = Q(t)\nabla^{x}\mathbf{v}(\mathbf{x},t)Q(t)^{T} + \dot{Q}(t)Q(t)^{T}.$$

The rate-of-strain tensors satisfy

$$L^*(\mathbf{x}^*,t)|_{x^*=g(x,t)} = Q(t)L(\mathbf{x},t)Q(t)^{T}.$$



Proof.

Recall the left and right polar decompositions

$$F = VR = RU,$$

where R is a rotation and

$$U = \sqrt{F^{\mathsf{T}}F}, \qquad V = \sqrt{FF^{\mathsf{T}}}, \qquad C = F^{\mathsf{T}}F.$$

Taking the $\mathbf{X}$-gradient of $\boldsymbol{\phi}^*$ gives

$$\begin{aligned} F^*(\mathbf{X},t) &= \nabla^{\mathsf{X}}\boldsymbol{\phi}^*(\mathbf{X},t) \\ &= \nabla^{\mathsf{X}}\left(Q(t)\boldsymbol{\phi}(\mathbf{X},t)+\mathbf{c}(t)\right) \\ &= Q(t)\nabla^{\mathsf{X}}\boldsymbol{\phi}(\mathbf{X},t) \\ &= Q(t)F(\mathbf{X},t). \end{aligned}$$

Proof (Cont.)

Now,

$$\begin{aligned} U^* &= \sqrt{F^{*\mathsf{T}} F^*} \\ &= \sqrt{F^{\mathsf{T}} Q^{\mathsf{T}} Q F} \\ &= \sqrt{F^{\mathsf{T}} F} \\ &= U. \end{aligned}$$

Also,

$$R^* U^* = F^* = QF = QRU.$$

Therefore,

$$R^* U = QRU$$

gives two polar decompositions of F^*. Both R^* and QR are rotation tensors. By uniqueness of the polar decomposition,

$$R^* = QR.$$

Proof (Cont.)

The remaining material identities follow directly:

$$C^* = F^{*\mathsf{T}} F^* = F^{\mathsf{T}} Q^{\mathsf{T}} Q F = C,$$
$$V^* = F^* R^{*\mathsf{T}} = Q F R^{\mathsf{T}} Q^{\mathsf{T}} = Q V Q^{\mathsf{T}}.$$

Differentiating equation (5.36) with respect to time and using the chain rule gives the displayed transformation of the spatial velocity gradient. Since $\dot{Q} Q^{\mathsf{T}}$ is skew-symmetric, taking symmetric parts yields

$$L^* = Q L Q^{\mathsf{T}}.$$

The componentwise details are left as an exercise. □

Definition (5.23 (Frame-Indifferent Fields))

The scalar field ϕ, vector field $\mathbf{w}$, and second-order tensor field S are called *frame-indifferent* if, for every superposed rigid motion $\mathbf{x}^* = \mathbf{g}(\mathbf{x}, t) = Q(t)\mathbf{x} + \mathbf{c}(t)$,

$$\phi^*(\mathbf{x}^*, t) = \phi(\mathbf{x}, t),$$

$$\mathbf{w}^*(\mathbf{x}^*, t) = Q(t)\mathbf{w}(\mathbf{x}, t),$$

and

$$S^*(\mathbf{x}^*, t) = Q(t)S(\mathbf{x}, t)Q(t)^{\mathsf{T}},$$

for every $\mathbf{x} \in B_t$ and $t \geq 0$.



Axiom 5.24 (Frame Indifference)

The spatial mass density $\rho(\mathbf{x}, t)$, temperature $\theta(\mathbf{x}, t)$, entropy per unit mass $\eta(\mathbf{x}, t)$, internal energy per unit mass $\phi(\mathbf{x}, t)$, Cauchy stress $S(\mathbf{x}, t)$, and Fourier heat flux $\mathbf{q}(\mathbf{x}, t)$ are all frame-indifferent fields.

Proposition (Restrictions on Constitutive Functions)

Suppose that the internal energy, heat flux, and Cauchy stress are modeled by

$$\phi = \widehat{\phi}(\rho, \theta, L), \qquad \mathbf{q} = \widehat{\mathbf{q}}(\rho, \theta, L), \qquad S = \widehat{S}(\rho, \theta, L),$$

where $\widehat{\phi}$, $\widehat{\mathbf{q}}$, *and* $\widehat{S}$ *are functions. If Axiom 5.24 holds, then*

$$\widehat{\phi}(\rho, \theta, QLQ^{\mathsf{T}}) = \widehat{\phi}(\rho, \theta, L),$$

$$\widehat{\mathbf{q}}(\rho, \theta, QLQ^{\mathsf{T}}) = Q\widehat{\mathbf{q}}(\rho, \theta, L),$$

and

$$\widehat{S}(\rho, \theta, QLQ^{\mathsf{T}}) = Q\widehat{S}(\rho, \theta, L)Q^{\mathsf{T}},$$

for every rotation Q *and every admissible* ρ, θ, *and* L.



Proof.

By Axiom 5.24 and Definition 5.23,

$$\widehat{\phi}(\rho^*, \theta^*, L^*) = \widehat{\phi}(\rho, \theta, L),$$

$$\widehat{\mathbf{q}}(\rho^*, \theta^*, L^*) = Q\widehat{\mathbf{q}}(\rho, \theta, L),$$

and

$$\widehat{S}(\rho^*, \theta^*, L^*) = Q\widehat{S}(\rho, \theta, L)Q^{\mathsf{T}}.$$

Since $\rho^* = \rho$ and $\theta^* = \theta$ by Axiom 5.24, and $L^* = QLQ^{\mathsf{T}}$ by Result 5.22, the stated restrictions follow. □

Frame indifference therefore imposes severe restrictions on the constitutive relations used to close continuum models.



Material Constraints

Definition (5.25 (Material Constraint))

A continuum body is subject to a *simple material constraint*, or *internal constraint*, if every motion

$$\boldsymbol{\phi} : B \times [0, \infty) \longrightarrow \mathbb{E}^3$$

must satisfy an equation of the form

$$\gamma\big(F(\mathbf{X}, t)\big) = 0, \qquad \mathbf{X} \in B, \quad t \geq 0, \tag{5.37}$$

where F is the deformation gradient and

$$\gamma : \mathcal{V}^2 \longrightarrow \mathbb{R}$$

is a prescribed function.

Example (Incompressibility)

A material that admits no volume change is called *incompressible*. In view of Result 4.12, an appropriate constraint is

$$\gamma(F) = \det F - 1. \tag{5.38}$$

When $\gamma(F) = 0$,

$$\det F(\mathbf{X}, t) = 1, \qquad \mathbf{X} \in B, \quad t \geq 0.$$

An equivalent spatial form of the constraint is

$$\nabla^{\mathrm{x}} \cdot \mathbf{v}(\mathbf{x}, t) = 0, \qquad \mathbf{x} \in B_t, \quad t \geq 0.$$

Axiom 5.26 (Active and Reactive Stress)

The first Piola–Kirchhoff stress field in a continuum body subject to the material constraint $\gamma(F) = 0$ can be decomposed as

$$P(\mathbf{X}, t) = P^{(r)}(\mathbf{X}, t) + P^{(a)}(\mathbf{X}, t),$$

where $P^{(a)}$ is an *active stress field* determined by a constitutive equation, and $P^{(r)}$ is a *reactive stress field* whose stress power is zero in every possible motion:

$$P^{(r)} : \dot{F} = 0, \qquad \mathbf{X} \in B, \quad t > 0. \tag{5.39}$$

Proposition (Form of the Reactive Piola Stress)

Suppose Axiom 5.26 holds and $\gamma(F) = 0$ *is an internal constraint. Then at every regular constrained deformation, where* $D\gamma(F) \neq 0$*, the most general form of the reactive stress field is*

$$P^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) D\gamma\big(F(\mathbf{X}, t)\big), \qquad \mathbf{X} \in B, \quad t > 0,$$

where $q(\mathbf{X}, t)$ *is an unknown scalar field called a multiplier.*

For incompressibility, equation (5.38) *gives*

$$P^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) \det F(\mathbf{X}, t) F(\mathbf{X}, t)^{-\mathrm{T}}. \tag{5.40}$$



Proof.

Equation (5.39) gives

$$P^{(r)} : \dot{F} = 0, \qquad \mathbf{X} \in B, \quad t > 0.$$

Because the constraint $\gamma(F) = 0$ holds at all times,

$$\begin{aligned} 0 &= \frac{\mathrm{d}}{\mathrm{d}t}\gamma\big(F(\mathbf{X}, t)\big) \\ &= D\gamma\big(F(\mathbf{X}, t)\big) : \dot{F}(\mathbf{X}, t), \end{aligned}$$

for every $\mathbf{X} \in B$ and $t > 0$.

At a regular constrained deformation, the admissible rates span the tangent space

$$T_F\mathcal{C} = \left\{ H \in \mathcal{V}^2 \;\middle|\; D\gamma(F) : H = 0 \right\}.$$



Proof (Cont.)

Both $P^{(r)}$ and $D\gamma(F)$ are orthogonal to this tangent space. Its orthogonal complement is one-dimensional, so these tensors are parallel:

$$P^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) D\gamma\big(F(\mathbf{X}, t)\big),$$

where q is an undetermined multiplier, not a constitutively prescribed function of F.

Proof (Cont.)

For the incompressibility constraint

$$\gamma(F) = \det F - 1,$$

Result 2.22 gives

$$D\gamma(F) = \det(F)F^{-\mathrm{T}}.$$

Therefore,

$$P^{(r)}(\mathbf{X}, t) = q(\mathbf{X}, t) \det F(\mathbf{X}, t)F(\mathbf{X}, t)^{-\mathrm{T}},$$

which is equation (5.40). □

Proposition (Reactive Cauchy Stress)

Suppose Axiom 5.26 holds and $\gamma(F) = 0$ is a material constraint. The Cauchy stress may be decomposed as

$$S(\mathbf{x}, t) = S^{(r)}(\mathbf{x}, t) + S^{(a)}(\mathbf{x}, t),$$

where $S^{(r)}$ corresponds to $P^{(r)}$ and $S^{(a)}$ corresponds to $P^{(a)}$.

For an incompressible material, the most general reactive Cauchy stress is

$$S^{(r)}(\mathbf{x}, t) = p(\mathbf{x}, t) I,$$

where

$$p(\mathbf{x}, t) = q_s(\mathbf{x}, t)$$

is the spatial description of the material multiplier q and is usually called the pressure.

Proof.

Exercise. □



Isothermal Considerations



Many continuum processes are almost purely mechanical and can be approximated as isothermal.

Definition (5.27 (Isothermal Process))

By a *thermo-mechanical process* for a continuum body, we mean a pair of functions $(\boldsymbol{\phi}, \Theta)$, where $\boldsymbol{\phi}$ is an admissible motion and $\Theta(\mathbf{X}, t)$ is a material temperature field.

The process is called *isothermal* if Θ is constant, independent of $\mathbf{X}$ and t. Equivalently, the spatial temperature field $\theta(\mathbf{x}, t)$ is constant, independent of $\mathbf{x}$ and t.



Remark (Balance Laws in an Isothermal Model)

For isothermal processes, the internal-energy balance is typically omitted, because it directly relates the heat flux, heat supply, and internal energy. The heat flux and heat supply are usually taken to be zero in an isothermal model.

Ironically, the Clausius–Duhem inequality remains important in isothermal modeling.



Definition (5.28 (Closed Process))

A thermo-mechanical process $(\boldsymbol{\phi}, \Theta)$ for a body is said to be *closed* on an interval $[t_0, t_1]$ if, for every $\mathbf{X} \in B$,

$$\boldsymbol{\phi}(\mathbf{X}, t_0) = \boldsymbol{\phi}(\mathbf{X}, t_1),$$

$$\dot{\boldsymbol{\phi}}(\mathbf{X}, t_0) = \dot{\boldsymbol{\phi}}(\mathbf{X}, t_1),$$

and

$$\Theta(\mathbf{X}, t_0) = \Theta(\mathbf{X}, t_1).$$

A body is called *energetically passive* if, for every closed process, its material free-energy density satisfies

$$\Psi(\mathbf{X}, t_1) \geq \Psi(\mathbf{X}, t_0), \qquad \mathbf{X} \in B.$$

Result 5.29 (Mechanical Energy Inequality)

Consider an energetically passive body with reference configuration B. Then, for any closed isothermal process on an interval $[t_0, t_1]$,

$$\int_{t_0}^{t_1} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t) \, \mathrm{d}t \geq 0, \qquad \mathbf{X} \in B. \tag{5.41}$$

Here $P(\mathbf{X}, t)$ is the first Piola–Kirchhoff stress field and $F(\mathbf{X}, t)$ is the deformation gradient. Equivalently,

$$\int_{t_0}^{t_1} \det F(\mathbf{X}, t) \, S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t) \, \mathrm{d}t \geq 0, \qquad \mathbf{X} \in B, \tag{5.42}$$

Here

$$S_m(\mathbf{X}, t) := S\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big), \qquad L_m(\mathbf{X}, t) := L\big(\boldsymbol{\phi}(\mathbf{X}, t), t\big)$$

are the material descriptions of the Cauchy stress and rate-of-strain fields.

Proof.

For any thermo-mechanical process, the reduced Clausius–Duhem inequality in Lagrangian form, Result 5.20, gives

$$\rho_0 \dot{\psi} = P : \dot{F} - \rho_0 \eta_m \dot{\Theta} - \Theta^{-1} \mathbf{Q} \cdot \nabla^X \Theta - \Theta \sigma_m \det F .$$

Since $\Theta \sigma_m \det F \geq 0$, it follows that

$$\rho_0 \dot{\psi} \leq P : \dot{F} - \rho_0 \eta_m \dot{\Theta} - \Theta^{-1} \mathbf{Q} \cdot \nabla^X \Theta, \qquad \mathbf{X} \in B, \quad t \geq 0.$$

If the process is isothermal, then

$$\dot{\Theta} = 0 \qquad \text{and} \qquad \nabla^X \Theta = \mathbf{0}.$$

Consequently,

$$\rho_0(\mathbf{X}) \dot{\psi}(\mathbf{X}, t) \leq P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t), \qquad \mathbf{X} \in B, \quad t \geq 0.$$

Proof (Cont.)

Integrating over the time interval $[t_0, t_1]$ gives

$$\int_{t_0}^{t_1} \rho_0(\mathbf{X})\dot{\Psi}(\mathbf{X}, t)\, \mathrm{d}t \leq \int_{t_0}^{t_1} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t)\, \mathrm{d}t,$$

$$\rho_0(\mathbf{X})\big[\Psi(\mathbf{X}, t_1) - \Psi(\mathbf{X}, t_0)\big] \leq \int_{t_0}^{t_1} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t)\, \mathrm{d}t.$$

Because the process is closed and the body is energetically passive,

$$\Psi(\mathbf{X}, t_1) - \Psi(\mathbf{X}, t_0) \geq 0.$$

Since $\rho_0(\mathbf{X}) > 0$, the left-hand side above is nonnegative. Hence

$$\int_{t_0}^{t_1} P(\mathbf{X}, t) : \dot{F}(\mathbf{X}, t)\, \mathrm{d}t \geq 0,$$

which is (5.41).

Proof (Cont.)

Similarly, for any process the reduced Clausius–Duhem inequality in Eulerian form, Result 5.12, states

$$\rho\dot{\psi} \leq S : L - \rho\eta\dot{\theta} - \theta^{-1}\mathbf{q}\cdot\nabla^{\mathrm{x}}\theta, \qquad \mathbf{x}\in B_t, \quad t\geq 0.$$

If the process is isothermal, then

$$\dot{\theta} = 0 \qquad \text{and} \qquad \nabla^{\mathrm{x}}\theta = \mathbf{0},$$

and therefore

$$\rho(\mathbf{x},t)\dot{\psi}(\mathbf{x},t) \leq S(\mathbf{x},t) : L(\mathbf{x},t), \qquad \mathbf{x}\in B_t, \quad t\geq 0.$$

In material form, this reads

$$\rho_m(\mathbf{X},t)\dot{\Psi}(\mathbf{X},t) \leq S_m(\mathbf{X},t) : L_m(\mathbf{X},t), \qquad \mathbf{X}\in B, \quad t\geq 0,$$

where $\Psi = \psi_m$.

Proof (Cont.)

By conservation of mass, Result 5.13,

$$\rho_m(\mathbf{X}, t) = \frac{\rho_0(\mathbf{X})}{\det F(\mathbf{X}, t)}.$$

Substitution into the material inequality gives

$$\rho_0(\mathbf{X})\dot{\Psi}(\mathbf{X}, t) \leq \det F(\mathbf{X}, t)\, S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t).$$

Proof (Cont.)

Integrating over $[t_0, t_1]$ yields

$$\rho_0(\mathbf{X})\big[\Psi(\mathbf{X}, t_1) - \Psi(\mathbf{X}, t_0)\big] \leq \int_{t_0}^{t_1} \det F(\mathbf{X}, t) \cdot S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t)\, \mathrm{d}t.$$

The left-hand side is nonnegative because the process is closed and the body is energetically passive. Thus

$$\int_{t_0}^{t_1} \det F(\mathbf{X}, t)\, S_m(\mathbf{X}, t) : L_m(\mathbf{X}, t)\, \mathrm{d}t \geq 0,$$

which is (5.42). □